

Sophia Tang

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EDUCATION

University of Pennsylvania

Philadelphia, PA

Dual Degree with Bachelor of Science in Engineering and Bachelor of Science in Economics

Sept 2023 - May 2027 (*expected*)

Jerome Fisher Program in Management & Technology; Major in Computer Science & Statistics

Relevant Coursework: Algorithms, Programming Languages and Techniques I and II, Mathematics of Computer Science, Multivariable Calculus, Linear Algebra, Machine Learning, Computer Systems, Automata Computability and Complexity, Corporate Finance, Database & Information Systems, Statistical Inference, Convex Optimization.

RESEARCH EXPERIENCE

Kempner Institute, Harvard University.

Boston, MA

Visiting Undergraduate Researcher. **Advised by Dr. Michael Albergo.**

May 2026 - Present

- Developing theoretical generative modeling frameworks.

Chatterjee Lab, University of Pennsylvania.

Philadelphia, PA

Undergraduate Researcher. **Advised by Dr. Pranam Chatterjee.**

Sept 2024 - Present

- Developing theoretical ML frameworks for generative modeling with applications in biological design and simulation.
- **First-author papers published at ICML 2025 and ICLR 2026, and several papers under review (see below).**

Mitchell Lab, University of Pennsylvania.

Philadelphia, PA

Undergraduate Researcher. **Advised by Dr. Michael Mitchell.**

Sept 2023 - Present

- Developed novel targeted lipid nanoparticle (LNP)-encapsulated mRNA therapeutics with high transport across the blood-brain barrier after systemic administration and transfection into brain cell types.
- **Several published co-authored papers and a first-author review paper (see below).**

MACHINE LEARNING PUBLICATIONS (*denotes equal contribution; † denotes advising role)

[1] PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion.

Sophia Tang*, Yinuo Zhang*, Pranam Chatterjee†.

International Conference on Machine Learning (ICML 2025), Dec 2024. <https://openreview.net/forum?id=FQoy1Y1Hd8>

Description: We introduce PepTune, a multi-objective discrete diffusion model for simultaneous generation and optimization of therapeutic peptide SMILES. Built on the Masked Discrete Language Model (MDLM) framework, PepTune ensures valid peptide structures with a novel bond-dependent masking schedule and invalid loss function. To guide the diffusion process, we introduce Monte Carlo Tree Guidance (MCTG), an inference-time multi-objective guidance algorithm that balances exploration and exploitation to iteratively refine Pareto-optimal sequences. MCTG integrates classifier-based rewards with search-tree expansion, overcoming gradient estimation challenges and data sparsity.

[2] Branched Schrödinger Bridge Matching.

Sophia Tang, Yinuo Zhang, Alexander Tong†, Pranam Chatterjee†.

International Conference on Learning Representations (ICLR 2026). Jun 2025. <https://doi.org/10.48550/arXiv.2506.09007>

Description: We introduce Branched Schrödinger Bridge Matching (BranchSBM), a novel framework that learns branched Schrödinger bridges. BranchSBM parameterizes multiple time-dependent velocity fields and growth processes, enabling the representation of population-level divergence into multiple terminal distributions. We show that BranchSBM is not only more expressive but also essential for tasks involving multi-path surface navigation, modeling cell fate bifurcations from homogeneous progenitor states, and simulating diverging cellular responses to perturbations.

[3] Foundations of Schrödinger Bridges for Generative Modeling

Sophia Tang

Preprint. Mar 2026. <https://doi.org/10.48550/arXiv.2603.18992>

Description: This book (220 pages, 24 figures) covers the mathematical foundation of Schrödinger bridges as a unifying principle underlying modern generative modeling techniques.

[4] TR2-D2: Tree Search Guided Trajectory-Aware Fine-Tuning for Discrete Diffusion.

Sophia Tang*, Yuchen Zhu*, Molei Tao†, Pranam Chatterjee†.

Preprint. Sept 2025. <https://doi.org/10.48550/arXiv.2509.25171>

Description: We introduce a novel framework that optimizes reward-guided discrete diffusion trajectories with tree search to construct replay buffers for trajectory-aware fine-tuning.

[5] Entangled Schrödinger Bridge Matching.

Sophia Tang, Yinuo Zhang, Pranam Chatterjee†.

Preprint. Oct 2025. <https://doi.org/10.48550/arXiv.2511.07406>

Description: We introduce a framework that learns the first- and second-order stochastic dynamics of interacting, multi-particle systems where the direction and magnitude of each particle's path depend dynamically on the paths of the other particles.

[6] A2D2: Finetuning Any-Length Discrete Diffusion for Adaptive Decoding.

Sophia Tang, Yuchen Zhu, Molei Tao†, Pranam Chatterjee†.

Preprint. April 2026. <https://openreview.net/forum?id=KDufogRwN>

[7] Gumbel-Softmax Flow Matching with Straight-Through Guidance for Controllable Biological Sequence Generation.

Sophia Tang, Yinnuo Zhang, Alexander Tong†, Pranam Chatterjee†.

Preprint. Mar 2025. <https://doi.org/10.48550/arXiv.2503.17361>

[8] mRNAAutilus: Multi-Objective-Guided Discrete Generation of mRNA with Optimized Therapeutic Properties.

Savan Patel*, Sophia Tang*, Yesol Kim*, Yinnuo Zhang, Divya Srijay, Ping-Jung Lin, Shambhavi Shubham, Fengmei Pi, Cedric Wu, Sherwood Yao†, and Pranam Chatterjee†.

Preprint. Jun 2026. <https://doi.org/10.48550/arXiv.2605.31296>

[9] PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction.

Yinnuo Zhang, Sophia Tang, Tong Chen, Elizabeth Mabood, Sophia Vincoff and Pranam Chatterjee†.

Nature Communications (In Print). Jan 2026. <https://www.biorxiv.org/content/10.64898/2025.12.31.697180v1>

[10] TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation.

Hanqun Cao*, Aastha Pat†, Sophia Tang, Yinnuo Zhang, Jingjie Zhang, Pheng Ann Heng, Pranam Chatterjee†

International Conference on Machine Learning (ICML 2026). May 2026. <https://arxiv.org/abs/2605.09810>

[11] Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design.

Tong Chen, Yinnuo Zhang, Sophia Tang, and Pranam Chatterjee†.

Preprint. May 2025. <https://doi.org/10.48550/arXiv.2505.07086>

BIOENGINEERING PUBLICATIONS († denotes advising role)

[12] Peptide-functionalized nanoparticles for brain-targeted therapeutics.

Sophia Tang, Emily L. Han, and Michael J. Mitchell†.

Drug Delivery and Translational Research, Springer Nature. Mar 2025. <https://doi.org/10.1007/s13346-025-01840-w>

[13] Peptide-Functionalized Lipid Nanoparticles for Targeted Systemic mRNA Delivery to the Brain.

Emily L. Han, Sophia Tang, Dongyoon Kim, Amanda M. Murray, Kelsey L. Swingle, Alex G. Hamilton, Kaitlin Mrksich, Marshall S. Padilla, Rohan Palanki, Jacqueline J. Li, and Michael J. Mitchell†.

ACS Nano Letters. Dec 2024. <https://doi.org/10.1021/acs.nanolett.4c05186>

[14] Predictive High-Throughput Platform for Dual Screening of mRNA Lipid Nanoparticle Blood–Brain Barrier Transfection and Crossing.

Emily L. Han, Marshall S. Padilla, Rohan Palanki, Dongyoon Kim, Kaitlin Mrksich, Jacqueline J. Li, Sophia Tang, Il-Chul Yoon, and Michael J. Mitchell†.

ACS Nano Letters. Jan 2024. <https://doi.org/10.1021/acs.nanolett.3c03509>

[15] Optimized microfluidic formulation and organic excipients for improved lipid nanoparticle-mediated genome editing.

Rohan Palanki, Emily L. Han, Amanda M. Murray, Robin Maganti, Sophia Tang, Kelsey L. Swingle, Dongyoon Kim, Hannah Yamagata, Hannah C. Safford, Kaitlin Mrksich, William H. Peranteau, and Michael J. Mitchell†.

Lab on a Chip, Royal Society of Chemistry. Jan 2024. <https://doi.org/10.1039/D4LC00283K>

[16] High-Throughput In Vivo Screening Identifies Structural Factors Driving mRNA Lipid Nanoparticle Delivery to the Brain.

Emily L. Han, Dongyoon Kim, Amanda M. Murray, Kaitlin Mrksich, Alex G. Hamilton, Sophia Tang, Soyeon Yoo, Audrey T. Zhu, Elaine R. Tong, Rohan Palanki, Michelle L. Hall, Sean K. Bedingfield, Michael J. Mitchell†.

ACS Nano. Jan 2026. <https://doi.org/10.1021/acsnano.5c19138>

TECHNICAL ARTICLES

A Complete Guide to Spherical Equivariant Graph Transformers. <https://doi.org/10.48550/arXiv.2512.13927>

A 99-page breakdown of the theory behind spherical equivariant (SE(3)) graph neural networks (EGNNs).

A Complete Guide to Protein Folding Prediction. <https://alchemybio.substack.com/p/a-complete-guide-to-protein-folding>

A 3-hour breakdown of the three-track RoseTTAFold protein prediction model that leverages multi-modal deep learning architectures.

INVITED TALKS

Discrete Diffusion Reading Group. Nov 2025. https://youtu.be/NX2Mth_Epvk?si=LsVS_bhzyD-RV03m

Starkly Speaking Reading Group. Jul 2025. <https://youtu.be/inVYA0pQ4Wg?si=mdwhTE3wu0vXJ4ZX>

Learning on Graphs and Geometry Reading Group. Jan 2025. <https://youtu.be/KVr8ryclwdA?si=RdRPdqNqymbk3Og>

AWARDS AND HONOURS

CRA Outstanding Undergraduate Researcher Award. Jan 2026. 1 of 8 awardees across North America. The award is widely regarded as the highest honor for undergraduate computer science students and recognizes exceptional research accomplishments in computing.

<https://www.seas.upenn.edu/stories/sophia-tang-wins-cra-outstanding-undergraduate-research-award/>